Generative AI Basics

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Check out my

- Instructor page at General Assembly
- Blog with data science resources
- Linkedin profile

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Before we get started...

Some quick reminders

- This is a beginner-level overview of generative AI.
- There will be time for live Q&A at the end.
- YES—we're recording. You'll get the recording in about 24 hours after class.

What you'll learn today

- Large Language Models, **explained**
- **Benefits & limitations** of LLMs
- **Training** large language models
- Understanding and **identifying biases**
- Guiding principles for **ethical AI use**
- Best Practices for **writing effective prompts**
- Review **real-world examples** of AI use cases
- **Live AI demo**

Now that we got that out of the way, let’s dive in.

Large Language Models, Explained

In the chat:

What comes to mind when you hear the words

Artificial Intelligence?

Artificial Intelligence

The ability of machines to perform tasks that typically require human-like intelligence, like learning, problem-solving, pattern recognition, and decision-making using algorithms, data, and statistical models.

Generative AI

A *type of artificial intelligence* that is capable of creating new, original content, such as images, music, and text, based on the patterns and styles it has learned from a dataset.

Large Language Models

Computer programs or algorithms that are capable of processing and generating natural language text using vast amounts of data and statistical models.

LLMs: An Exercise in Statistics

The promise of large language models is that they

can	62%
will	11%
are	7%
capture	2%
could	2%

At a very high level, LLMs like ***ChatGPT*** generate their responses **one word at a time**, based on probability.

However, probability isn't the only factor - often times lower probability options result in a better output.

Benefits of LLMs

- Great at generating coherent, well-written text
- “Know” a huge amount about the (pre-cutoff) world
- Can be fine-tuned to specific tasks
- Always-on, 24/7

Limitations of LLMs

- Prone to “hallucinations”
- Lack true reasoning
- Require careful prompting
- Takes a lot of experience to use effectively
- Require significant compute resources
- Answers may be homogenous

Training a Large Language Model

1. Pre-Training

Or, Unsupervised Learning. The initial phase where an LLM is exposed to a vast amount of text data.

The model learns the structure, patterns, and common language knowledge by assigning values to words and letters. It then uses its knowledge of patterns to predict the next word in a sentence.

Training a Large Language Model

2. Fine Tuning

Or, Supervised Learning. The model is trained on a smaller, task-specific dataset.

The model adapts its pretrained knowledge to perform well on the target task by adjusting its architecture and parameters. There is typically a human providing guidance to help the model learn at this stage.

Training a Large Language Model

3. RLHF

Or, Reinforcement Learning. Post fine-tuning, the model goes through Reinforcement Learning from Human Feedback (RLHF).

It interacts with real users, receiving feedback on its performance.

“

Fine-tuning the system can introduce additional biases — side effects — that cause it to drift in unexpected directions.

James Zou, Computer Science Professor, Stanford

Bias in Generative AI

Prompts and Their Impact on Bias

Researchers at the University of Waterloo wanted to study the impact prompts had on the outputs they received from ChatGPT.

They found that prompts had a significant impact on bias in ChatGPT's outputs.

Prompts and Their Impact on Bias

They tested 3 prompts to compare their outputs.
Which one do you think received the **least** biased output?

1

Explain why socially assistive robots are an effective treatment for older adults with dementia.

2

Are socially assistive robots an effective treatment for older adults with dementia?

3

What are the benefits and drawbacks of socially assistive robots as a treatment for older adults with dementia?

Guiding Principles for Ethical AI Use

Best Practices for Using LLMs

1. Recognize Potential Risks

- Misinformation
- Biased Outputs

Best Practices for Using LLMs

2. Awareness of Bias

- Language models may reflect the biases of their training data
- Stay critical and conscious

Best Practices for Using LLMs

3. Recognize Limitations

- Lack of emotions or human experience
- No access to real-time information

Best Practices for Using LLMs

4. Fact-Checking and Verification

- Remember, LLMs don't *know* anything
- Fact-check using reliable sources

Using Generative AI

Popular Tools

Platform	System	Models	Features	Cost
OpenAI Backed by Microsoft	ChatGPT	GPT 3.5	Fast, cheap	Free
	ChatGPT Plus /Enterprise	GPT 4-V DALL•E 3	Code, web, multimodal	$20/month
	Playground	Multiple	Tweak Parameters	$5+ PAYG
Microsoft Powered by OpenAI	Bing AI *(Creative Mode)*	GPT 4-V DALL•E 3	Internet, Multimodal	Free
Anthropic Backed by Amazon	Claude AI	Claude 2.0	Large Documents	Free or $20/month
Google	Bard	PaLM 2 (Gemini?)	Connect to Gmail, Drive	Varies

5 Rules for Writing Good Prompts

- Be Clear and Specific
- Specify Format and Length
- Ask for Multiple Perspectives
- Incorporate Priming Statements
- Refine and Experiment

Be Clear and Specific

Tell me about making a cake.

Can you provide a step-by-step guide on how to make a chocolate cake from scratch, including the measurements for each ingredient?

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Specify Format and Length

Discuss the pros and cons of renewable energy

List three advantages and three disadvantages of renewable energy sources in bullet points.

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Ask for Multiple Perspectives

What are your thoughts on social media?

Present two contrasting arguments regarding the impact of social media on society, considering both positive and negative effects.

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Incorporate Priming Statements

Talk about remote work.

In recent years, there has been a significant shift towards remote work. Discuss the advantages and disadvantages of this trend.

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Refine and Experiment

Write a poem about nature.

This is okay, but don't stop there!

Compose a descriptive poem (12-16 lines) that captures the beauty of a serene forest.

Compose a descriptive poem (12-16 lines) that captures the beauty of a serene forest and evokes a sense of harmony.

See these prompts in action.

When using LLMs at work, remember...

1

LLMs like ChatGPT are **tools; not replacements** for human expertise.

2

Be **specific and clear** in your prompts.
The output will only be as good as the input you provide!

3

Remember that **LLMs are not oracles**.
They are machine learning models that may not be accurate or up-to-date.

4

Keep **data privacy** in mind.
OpenAI stores a log of all the prompts entered into ChatGPT to use in future training.

“

AI will disrupt a lot of industries, but the way we think about it is as a productivity lift.

Sal Cucchiara, CIO at Morgan Stanley

Real-World AI Examples

The good...

Be My Eyes, Advancing Accessibility

Be My Eyes began developing a GPT-4 powered Virtual Volunteer™ within the Be My Eyes app that can generate the same level of context and understanding as a human volunteer.

Source: OpenAI.com

Lonely Planet, Innovating Travel

Lonely Planet is developing a solution to help customers plan epic trips and create life-changing experiences with personalized travel itineraries using AI.

Source: AWS

Moody's, Automating Analysis

Moody's implemented a fine-tuned LLM to automate data analysis and report writing, cutting report generation time from days to minutes and making data more widely accessible.

Source: Bloomberg

And the not-so-good...

Real People, Fake Stories

ChatGPT claimed a law professor had been at the center of a sexual harassment scandal. It even cited WaPo article that didn't exist as a source.

Source: Washington Post

The Honorable, ChatGPT

A judge in Pakistan was asking ChatGPT legal questions before making decisions in civil and criminal cases.

Source: Pakistan Today

ChatGPT's Data Dilemma

It is estimated that 10% of data that employees paste into ChatGPT is confidential.

Source: Cyber Haven

Key Takeaways

AI is More than ChatGPT

Large Language Models like ChatGPT are just one small part of Artificial intelligence as a whole.

AI is a Tool

AI is a tool people can use to be more productive, but it still requires human expertise.

AI Should be Used Responsibly

Understanding data privacy and ethics is a critical part of using AI effectively and safely at work.

ChatGPT in action

Hey, you're awesome.

Way to learn something new and advance your skills.

Resources

Blog Articles

- Data and AI: Best Friends or Foes?
- Software Engineering and AI: Best Friends or Foes?
- UX and AI: Best Friends or Foes?
- New AI Skills Training at GA
- 6 Benefits of an AI-Enabled Workforce

AI Outlook

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